

Lecture 6 Notes: Vector Space View of Linear Systems

1. Existence and Uniqueness for Linear Systems

[Likely exam topic] The lecture begins by separating two questions about a linear system $Ax = b$:

- **Existence:** Is there at least one solution?
- **Uniqueness:** If a solution exists, is it the only one, or are there many?

The instructor recalls the toy case of a 2×2 system. Each equation gives a line, and the possible geometric cases are:

- The lines intersect at one point: one unique solution.
- The lines are parallel and distinct: no solution.
- The lines lie on top of each other: infinitely many solutions.

The goal is to extend this picture from two equations and two unknowns to an arbitrary number of equations and unknowns.

2. Two Ways to View Matrix-Vector Multiplication

The instructor emphasizes that there are two key perceptions of Ax . This is described as a simple but important observation and a key to the later vector-space analysis.

2.1 Column View: Ax as a Linear Combination of Columns

If A is partitioned into columns, then multiplying A by x from the right forms a linear combination of the columns of A :

$$Ax = x_1c_1 + x_2c_2 + \cdots + x_nc_n.$$

From this point of view, the problem $Ax = b$ asks:

Can b be expressed as a linear combination of the columns of A ?

This column view is directly connected to the **existence** of solutions.

2.2 Row View: Ax as Inner Products with Rows

If A is partitioned into rows, then multiplying A by x forms a vector whose entries are inner products of x with the rows of A :

$$Ax = \begin{bmatrix} r_1^T x \\ r_2^T x \\ \vdots \\ r_m^T x \end{bmatrix}.$$

[Exam note] The instructor reminds students about his visual notation: a row vector such as r_1^T may be represented using a column-vector notation and then transposed.

This row view will be used later to understand **uniqueness**.

2.3 Transpose Mapping

Multiplying A^T from the right by a vector is equivalent, after transposing, to multiplying A from the left by a row vector. The instructor notes:

- $A^T x$ takes linear combinations of the columns of A^T .

- The columns of A^T are the rows of A .
- Therefore $A^T x$ can be interpreted as taking linear combinations of the rows of A .
- $A^T x$ can also be interpreted as taking inner products with the columns of A .

This A^T mapping will be used in the vector-space method for analyzing $Ax = b$.

3. A as a Linear Mapping

For an $m \times n$ matrix A , the product Ax maps

$$x \in \mathbb{R}^n \quad \longmapsto \quad Ax \in \mathbb{R}^m.$$

The instructor describes:

- The **input space** is $\mathbb{R}^n$.
- The **output space** is $\mathbb{R}^m$.
- b lies in the output space.
- The rows of A lie in the input space, because they take inner products with x .
- The columns of A lie in the output space.

He gives an informal brute-force interpretation: if one tried to solve $Ax = b$ by trial, one would vary x in the input space and watch the output Ax in $\mathbb{R}^m$, hoping it lands on the given b . The geometric question is therefore: which output vectors can this mapping actually generate?

All possible outputs generated by Ax are linear combinations of the columns of A . Therefore, all possible outputs form the span of the columns.

4. Column Space / Range Space of A

The first fundamental subspace introduced is the space spanned by the columns of A .

4.1 Definition

The **column space** of A is

$$\text{Col}(A) = \text{span}\{c_1, c_2, \dots, c_n\}.$$

The instructor also calls this the **range space** of A , because it is the set of all possible outputs:

$$\mathcal{R}(A) = \{Ax : x \in \mathbb{R}^n\}.$$

The transcript sometimes uses wording like “rank space” or “rate space”; in context, this is the range/column space.

4.2 Alternative Descriptions

The range space of A can be described in three equivalent ways:

1. The set of all possible outputs Ax generated by all possible $x \in \mathbb{R}^n$.
2. The span of the columns of A .
3. The set of all right-hand sides b for which $Ax = b$ has at least one solution.

5. Existence of Solutions

[Likely exam topic] The geometric conclusion about existence is:

$$Ax = b \text{ has a solution} \iff b \in \mathcal{R}(A) = \text{Col}(A).$$

If b lies outside the range/column space, then b cannot be generated by any Ax , so there is no solution.

If b lies inside the range/column space, then at least one x maps to b , so there is at least one solution.

5.1 Finite-Dimensional Setting

In response to a question, the instructor clarifies that the lecture is working with finite-dimensional spaces. Extensions to general operators are beyond the current level of discussion.

6. Guaranteeing Existence for Every b

The instructor then considers a stronger question:

For a given matrix A , can we guarantee a solution for every possible $b \in \mathbb{R}^m$?

This is not asking about one chosen b . It is classifying matrices A for which every right-hand side is solvable.

The condition is:

$$\mathcal{R}(A) = \mathbb{R}^m.$$

That means the column space covers the entire target/output space, so b has nowhere outside the column space to “hide.”

6.1 Necessary Shape Condition

Since the target space is m -dimensional and the range is spanned by the columns of A , there must be at least m columns available to span all of $\mathbb{R}^m$. Thus:

$$n \geq m.$$

So A must be either:

- square, or
- fat/wide.

This condition is necessary but not sufficient. Not every square or fat matrix has columns spanning $\mathbb{R}^m$.

For tall matrices, existence for every b cannot be guaranteed because there are not enough columns to cover the whole output space.

7. Right Inverses from Full Range Space

[Exam note] The instructor reviews how the right inverse conclusion is obtained.

Assume

$$\mathcal{R}(A) = \mathbb{R}^m.$$

Then $Ax = b$ has a solution for every $b \in \mathbb{R}^m$. In particular, it has solutions for every standard basis vector:

$$e_1, e_2, \dots, e_m.$$

Choose one solution d_i for each equation

$$Ad_i = e_i.$$

Place these solution vectors as columns of a matrix D :

$$D = [d_1 \ d_2 \ \dots \ d_m].$$

Then

$$AD = [Ad_1 \ Ad_2 \ \dots \ Ad_m] = [e_1 \ e_2 \ \dots \ e_m] = I_m.$$

So if the range space of A is all of $\mathbb{R}^m$, then A has a right inverse D satisfying

$$AD = I_m.$$

The instructor remarks that this property looks sophisticated, but it is a direct consequence of being able to solve $Ax = b$ for every b . If A is fat and the condition holds, there are infinitely many right inverses.

8. Full Range Space and Linear Independence Properties

[Exam note] The instructor mentions additional equivalent properties and leaves some proof directions to students.

If $\mathcal{R}(A) = \mathbb{R}^m$, then the columns of A contain a basis for $\mathbb{R}^m$. That means:

- We can find m columns of A that are linearly independent.
- These m columns form a basis for the output space.

One proof idea mentioned by the instructor is a pruning argument: if the columns span the whole m -dimensional target space and there are more than m columns, dependent columns can be removed without changing the span until a basis of m independent columns remains.

The instructor stresses a distinction:

- This does **not** say all columns of A are linearly independent.
- It says that some subset of m columns is linearly independent.
- If $n > m$, the full set of columns is necessarily linearly dependent because there are more vectors than the dimension of the output space.

Another implication is about rows:

If the range space covers the whole target space, then the rows of A are linearly independent.

This is interesting because a property stated in terms of the column space has a consequence for the rows.

9. Proof Idea: Row Independence and Full Column Space

The instructor develops the relationship between row independence and columns spanning the output space.

Let $\alpha \in \mathbb{R}^m$, where m is the number of rows and the dimension of the output space.

Left multiplication by α^T gives

$$\alpha^T A.$$

There are two ways to interpret this product:

1. As a linear combination of the rows of A .
2. As a row vector whose entries are inner products of α with the columns of A .

The rows of A are linearly independent exactly when

$$\alpha^T A = 0 \implies \alpha = 0.$$

Equivalently, the only vector orthogonal to all columns of A is the zero vector.

The instructor gives the geometric idea:

- If there were a nonzero vector α orthogonal to every column of A , then α could not lie in the range space of A .
- To see this, suppose $\alpha \in \mathcal{R}(A)$. Then

$$\alpha = \beta_1 c_1 + \beta_2 c_2 + \cdots + \beta_n c_n.$$

- Take inner product with α on both sides:

$$\alpha^T \alpha = \beta_1 \alpha^T c_1 + \beta_2 \alpha^T c_2 + \cdots + \beta_n \alpha^T c_n.$$

- Since α is orthogonal to every c_i , the right side is zero.
- Therefore $\alpha^T \alpha = 0$, which can happen only if $\alpha = 0$.

Thus a nonzero vector orthogonal to all columns cannot be in the range space. If the only such vector is zero, the columns span the whole output space.

This is the proof-by-contradiction/contrapositive logic the instructor wants students to notice:

- If the rows were linearly dependent, there would be a nonzero α with $\alpha^T A = 0$.
- That same condition says α is orthogonal to every column of A .
- If the range space were all of $\mathbb{R}^m$, then this nonzero α would have to lie in the range space.
- But a nonzero vector orthogonal to all columns cannot lie in their span, because taking the inner product with itself would force $\alpha^T \alpha = 0$.

So full range space forces row independence. Conversely, the instructor says the reverse implication can also be argued by contradiction, though the lecture does not spell out every detail.

10. Moving from Existence to Uniqueness

Up to this point, the lecture has focused on existence of solutions. The instructor then shifts to uniqueness.

The main duality is:

- Column space is related to existence.
- Row space is related to uniqueness.
- Existence for every b is guaranteed when the column space is the whole output space.
- Uniqueness, when a solution exists, is guaranteed when the row space is the whole input space.

11. Row Space of A

11.1 Definition

The **row space** of A is the span of the rows of A :

$$\text{Row}(A) = \text{span}\{r_1, r_2, \dots, r_m\}.$$

It is a subspace of the input space $\mathbb{R}^n$.

11.2 Row Space as Range Space of A^T

The rows of A are the columns of A^T . Therefore:

$$\text{Row}(A) = \text{Col}(A^T) = \mathcal{R}(A^T).$$

[Warning] Here $\mathcal{R}$ means **range**, not row. Thus the row space of A is the range space of A^T . The instructor calls this notation potentially confusing because the row space is being described with the same range-space symbol used earlier for column spaces.

The instructor warns that A^T is not necessarily the inverse of A . The map

$$A^T y$$

goes from the output space back into the input-space dimension, but in general it does not undo Ax . In other words, $A^T(Ax)$ is not generally equal to x .

12. Null Space of A

The instructor defines a set of input vectors that map to the origin:

$$\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}.$$

This is the **null space** of A .

12.1 Null Space Is a Vector Space

The instructor first calls it a null set and then notes it is actually a vector space.

If $x \in \mathcal{N}(A)$, then $Ax = 0$. For any scalar α ,

$$A(\alpha x) = \alpha Ax = \alpha 0 = 0.$$

So scaled versions stay in the null space.

If $x, y \in \mathcal{N}(A)$, then

$$A(x + y) = Ax + Ay = 0 + 0 = 0.$$

So sums stay in the null space.

Therefore $\mathcal{N}(A)$ is a subspace of the input space.

12.2 Null Space as Orthogonal to the Row Space

Using the row interpretation of Ax , the equation $Ax = 0$ means every inner product between x and a row of A is zero.

Thus:

$$x \in \mathcal{N}(A) \iff x \text{ is orthogonal to every row of } A.$$

Therefore the null space is the space orthogonal to the row space:

$$\mathcal{N}(A) = \text{Row}(A)^\perp.$$

The instructor illustrates this with a three-dimensional picture:

- The row space can be represented as a plane.
- The null space can be represented as the perpendicular line.
- Both are subspaces of $\mathbb{R}^n$.

13. Why Null Space Controls Uniqueness

Suppose x is a solution of $Ax = b$:

$$Ax = b.$$

Let $z \in \mathcal{N}(A)$. Then $Az = 0$. Define

$$x' = x + z.$$

Then

$$Ax' = A(x + z) = Ax + Az = b + 0 = b.$$

So x' is also a solution.

If $z \neq 0$, then $x' \neq x$, giving at least two different solutions. Therefore, for uniqueness, there cannot be any nonzero vector in the null space.

The uniqueness condition is:

$$\mathcal{N}(A) = \{0\}.$$

This is called the **trivial null space**. The instructor notes that the null space can never be empty because 0 always maps to 0. The trivial null space is the smallest possible null space.

The instructor also frames this through the special right-hand side $b = 0$: if multiple input vectors map to the origin, then even for output 0 one cannot identify a unique generating input. The condition $\mathcal{N}(A) = \{0\}$ is therefore not only necessary but also sufficient for uniqueness whenever a solution exists.

14. Row Space Covering the Input Space

Since the null space is orthogonal to the row space, having only the zero vector in the null space means there is no nonzero direction orthogonal to all rows.

This implies the rows span the whole input space:

$$\text{Row}(A) = \mathbb{R}^n.$$

Thus the uniqueness condition can be stated equivalently as:

$$Ax = b \text{ has at most one solution for each } b \iff \mathcal{N}(A) = \{0\} \iff \text{Row}(A) = \mathbb{R}^n.$$

The instructor frames this as the dual of the existence condition:

- Existence for every b : $\text{Col}(A) = \mathbb{R}^m$.
- Uniqueness: $\text{Row}(A) = \mathbb{R}^n$.

15. Direct Sum, Disjoint Subspaces, and a Warning About Union

The instructor warns not to write

$$\mathbb{R}^n = \text{Row}(A) \cup \mathcal{N}(A).$$

The union of the row space and null space is not the whole ambient space in general. In the plane-line picture, the union would only include those two subspaces, not every vector in $\mathbb{R}^n$.

Instead, one uses a **direct sum** expression:

$$\mathbb{R}^n = \text{Row}(A) \oplus \mathcal{N}(A).$$

This means every vector in $\mathbb{R}^n$ can be written as

$$x = x_1 + x_2,$$

where

$$x_1 \in \text{Row}(A), \quad x_2 \in \mathcal{N}(A).$$

15.1 Disjoint Subspaces

Two subspaces are called **disjoint** in this lecture if their intersection is only the origin:

$$S_1 \cap S_2 = \{0\}.$$

Every subspace contains the origin, so disjoint subspaces still share 0.

The row space $\mathcal{R}(A^T)$ and the null space $\mathcal{N}(A)$ are disjoint in this sense. In this case they are also orthogonal complements.

The instructor notes that the definition of direct sum does not itself require orthogonality, but in the row-space/null-space case the two subspaces are orthogonal.

16. The First Three Fundamental Subspaces

At this stage, the lecture has defined three subspaces:

1. Column space / range space of A :

$$\mathcal{R}(A) = \text{Col}(A) \subseteq \mathbb{R}^m.$$

This is the span of the columns and determines existence.

2. Row space of A :

$$\text{Row}(A) = \mathcal{R}(A^T) \subseteq \mathbb{R}^n.$$

This is the span of the rows and is connected to uniqueness.

3. Null space of A :

$$\mathcal{N}(A) \subseteq \mathbb{R}^n.$$

This is the set of input vectors mapped to the origin and is orthogonal to the row space.

The instructor says the picture is still incomplete because we have described two subspaces in the input space but only one in the output space.

17. Null Space of A^T / Left Null Space

The missing output-space companion is the space of vectors orthogonal to all columns of A . This is the null space of A^T :

$$\mathcal{N}(A^T) = \{y \in \mathbb{R}^m : A^T y = 0\}.$$

Since $A^T y$ takes inner products of y with the columns of A , this is the set of all output-space vectors orthogonal to every column of A .

Thus:

$$\mathcal{N}(A^T) = \text{Col}(A)^\perp.$$

The row space of A^T is the column/range space of A , and the space orthogonal to it is $\mathcal{N}(A^T)$.

Therefore the output space can be decomposed as

$$\mathbb{R}^m = \mathcal{R}(A) \oplus \mathcal{N}(A^T).$$

In the instructor's picture, the extra "orange" output-space direction represents $\mathcal{N}(A^T)$, completing the four-subspace view: two subspaces in the input space and two in the output space.

The instructor explains the dimension intuition:

- If $\mathcal{R}(A)$ has dimension r , then its orthogonal complement $\mathcal{N}(A^T)$ has dimension $m - r$.
- Their dimensions add to the dimension m of the ambient output space.
- A basis for one together with a basis for the other spans the output space.

He notes that he has not given a complete proof here, but this is the intuition based on orthogonal complements.

17.1 Name: Left Null Space

$\mathcal{N}(A^T)$ is also called the **left null space** of A .

The reason is:

If $z \in \mathcal{N}(A^T)$, then

$$A^T z = 0.$$

Taking transposes gives

$$z^T A = 0^T.$$

So these are the vectors that annihilate A by left multiplication.

The instructor notes the naming convention:

- $\mathcal{N}(A)$ is usually just called the null space, not the right null space.
- $\mathcal{N}(A^T)$ is called the left null space of A .
- Kernel and null space refer to the same idea, but the phrase “left null space” specifically means the null space of A^T .

18. Example 1: A 3×3 Matrix with xy -Plane Column and Row Spaces

The instructor gives an easy-to-draw 3×3 example whose columns are:

$$c_1 = (1, 0, 0)^T, \quad c_2 = (0, 1, 0)^T, \quad c_3 = (0, 0, 0)^T.$$

The column space is the span of the first two coordinate vectors:

$$\text{Col}(A) = \text{span}\{(1, 0, 0)^T, (0, 1, 0)^T\},$$

which is the xy -plane.

In this particular example, the rows are also

$$(1, 0, 0), \quad (0, 1, 0), \quad (0, 0, 0),$$

so the row space is also the xy -plane. The instructor warns that this equality of row and column spaces is special to this example and not true in general.

18.1 Existence and Uniqueness in Example 1

Existence for every b is not guaranteed because the column space does not cover all of $\mathbb{R}^3$. Only b 's in the xy -plane are attainable.

Even when b lies in the column space, the solution is not unique because the null space is not trivial. In this example the null space is the whole z -axis direction.

19. Example 2: A 3×2 Tall Matrix

The second example is a 3×2 matrix mapping

$$\mathbb{R}^2 \rightarrow \mathbb{R}^3.$$

This is a tall matrix.

The instructor emphasizes:

- For tall matrices, existence for arbitrary $b \in \mathbb{R}^3$ cannot be guaranteed.
- The input dimension is lower than the output dimension, so the mapping cannot cover all of the output space.

In the example:

- The columns again span the xy -plane in $\mathbb{R}^3$.
- The row vectors include the two coordinate directions in the two-dimensional input space, and their span is the whole input space.
- Therefore the null space is just the origin.

19.1 Existence and Uniqueness in Example 2

For this tall matrix:

- A solution is not guaranteed for arbitrary b .
- If b lies in the range/column space, then there is a unique solution.

So such a matrix can have uniqueness for all solvable right-hand sides, while still failing to have existence for all right-hand sides.

20. Comparing Existence and Uniqueness Conditions

The lecture summarizes the dual conditions:

20.1 Existence for Every b

$$\mathcal{R}(A) = \mathbb{R}^m.$$

This requires the column space to cover the entire target/output space. A necessary shape condition is:

$$m \leq n.$$

So A must be square or fat. This is necessary, not sufficient.

When this condition holds, A has at least one right inverse:

$$AD = I_m.$$

20.2 Uniqueness

$$\mathcal{N}(A) = \{0\}.$$

Equivalently:

$$\text{Row}(A) = \mathbb{R}^n.$$

This requires enough rows to span the input space. A necessary shape condition is:

$$m \geq n.$$

So A must be square or tall. This is necessary, not sufficient.

When this condition holds, A has at least one left inverse.

The instructor obtains this by transpose duality: if $\mathcal{N}(A) = \{0\}$, then A^T satisfies the corresponding full-range/right-inverse condition; transposing that right-inverse relation gives a left inverse for A .

20.3 Both Existence for All b and Uniqueness

Both conditions require

$$m \leq n \quad \text{and} \quad m \geq n,$$

so both can only hold for square matrices.

For square matrices where both corresponding null spaces are trivial and both conditions hold, the right inverse and left inverse coincide. This common matrix is called the inverse of A , denoted A^{-1} .

The instructor also states:

- In the existence-for-all- b case, the rows are linearly independent.
- In the uniqueness case, the columns are linearly independent.
- These statements are related by transpose duality.

[Exam note] The instructor highlights the shape distinction: existence for every b needs square or fat matrices; uniqueness needs square or tall matrices; both together force square matrices.

21. Fundamental Result: Row Rank Equals Column Rank

[Exam note] The instructor introduces an important result before continuing the system-of-equations analysis.

Let

$$r_C = \dim(\text{Col}(A))$$

be the dimension of the column space in the output space, and let

$$r_R = \dim(\text{Row}(A))$$

be the dimension of the row space in the input space.

Although these subspaces live in different ambient spaces and are defined differently, their dimensions are equal:

$$r_C = r_R.$$

This common dimension is called the **rank** of A :

$$\text{rank}(A) = r_C = r_R.$$

The instructor notes that people may refer to column rank and row rank, but they are equal.

22. Proof Idea: Row Rank $\leq$ Column Rank

The instructor outlines one direction of the proof.

Let the row space have dimension r_R . Choose a basis for the row space:

$$v_1, v_2, \dots, v_{r_R}.$$

These vectors are linearly independent and span the row space.

Now map these basis vectors into the output space by multiplying by A :

$$Av_1, Av_2, \dots, Av_{r_R}.$$

Each Av_i lies in the column space $\text{Col}(A)$, because every output of A lies in the column space.

The key question is whether the set

$$\{Av_1, \dots, Av_{r_R}\}$$

is linearly independent.

Suppose a linear combination is zero:

$$\alpha_1 Av_1 + \alpha_2 Av_2 + \dots + \alpha_{r_R} Av_{r_R} = 0.$$

Factor out A :

$$A(\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_{r_R} v_{r_R}) = 0.$$

The vector

$$w = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_{r_R} v_{r_R}$$

lies in the row space because it is a linear combination of row-space basis vectors.

But $Aw = 0$, so w also lies in the null space.

The row space and null space intersect only at 0. Therefore:

$$w = 0.$$

Because the v_i 's are linearly independent, this implies

$$\alpha_1 = \alpha_2 = \dots = \alpha_{r_R} = 0.$$

Thus the vectors

$$Av_1, \dots, Av_{r_R}$$

are linearly independent. Since they lie in the column space, the column space must have dimension at least r_R :

$$r_R \leq r_C.$$

23. Reverse Direction and Rank

The instructor says the reverse argument is analogous:

- Choose a basis for the column space.
- Map it back using A^T .
- Show the corresponding independent set lies in the row space.

This gives

$$r_C \leq r_R.$$

Together:

$$r_R \leq r_C \quad \text{and} \quad r_C \leq r_R$$

so

$$r_R = r_C.$$

This shared dimension is the rank of A .

[Likely exam topic] Rank is the common dimension of the row space and the column space.

24. Instructor Remarks and Study Guidance

The instructor says this material is the core of the upcoming discussion of systems of linear equations. It is based on simple principles, but the class will build on these results.

He recommends that students who are not fully comfortable with the vector-space picture draw it on paper:

- Draw the range/column space.
- Draw the row space.
- Draw the null space.
- Draw the left null space.
- Write the main properties beside the picture.

[Exam note] The instructor says this is an important basis for the next lecture's conclusions about systems of linear equations.

The lecture ends with the note that the class will continue on Tuesday.

Source and Coverage Note

Source used: `corrected/lecture6_corrected.md` only.

Coverage: These notes preserve the lecture's chronological development: existence and uniqueness, column and row interpretations of Ax , range/column space, conditions for existence for all b , right inverses, row independence, row space, null space, direct sums, left null space, the two concrete examples, the duality between existence and uniqueness, and the proof idea that row rank equals column rank. Exam-marked comments and instructor warnings were folded into the relevant sections. No other lecture transcript was processed.